

SatQuery AI - Multimodal Remote Sensing Evidence Report

Autonomous Agentic Execution & Evidence Provenance Trace

Query:	"Highlight the water body referred to in the query."
Task Category:	GROUNDING
Specialist Tool:	RS Text-Guided Region Grounding Specialist (VRSBench)
Reliability Score:	94.1% (HIGH TRUST)
Execution Time:	60.3 ms

1. Agent Synthesis & Visual-Language Answer

Text-Guided Region Grounding Output (VRSBench Standard):

- Target Entity Grounded: 'Water Body / River Channel'.
- Bounding Box Coordinates [ymin, xmin, ymax, xmax]:
 - Normalized: [0.0996, 0.3496, 0.8984, 0.6484]
 - Pixel Grid: [51, 179, 460, 332]
- Region Spatial Extent: 13.67% of total image frame.

2. Auditable Agent Pipeline Execution Trace

- **[Perception Layer]** Format & CRS validated. Mode: Single Image (1 image(s)). Status: PASSED
- **[Query Engine]** Classified intent: 'GROUNDING'. Extracted entities: ['water']. Task: 'grounding'
- **[Mission Planner]** Selected tool: 'RS Text-Guided Region Grounding Specialist' with params {'iou_threshold': 0.5, 'target_entities': ['water']}. Generated 7 node task-graph.
- **[Specialist Tool Execution]** Executed specialist tool 'RS Text-Guided Region Grounding Specialist (VRSBench)' with model confidence 92.5%.
- **[Evidence Fusion]** Aligned 2 evidence streams. Spatial Bounding Box aligned with query subject.
- **[Trust & Uncertainty]** Calculated Reliability Score: 94.1% (HIGH TRUST). Cross-model agreement: 94.5%.
- **[Discovery Engine]** Autonomous scan completed: Autonomous scan identified 1 'Beyond Your Query' discovery item(s).
- **[Answer Generator]** Final evidence-linked response synthesized in 60.3 ms.